

DYNAMIC ARRAY FORMULAS

THE GLASSDOOR DATASET

	A	B	C	D	E
1	Industry	Job Title	City	Average Salary	YoY % Growth
2	Technology	Web Developer	Washington DC	\$89,420	2.6%
3	Technology	Web Developer	Seattle	\$89,770	2.4%
4	Technology	Web Developer	San Francisco	\$104,640	3.3%
5	Technology	Web Developer	Philadelphia	\$74,319	3.6%
6	Technology	Web Developer	New York City	\$87,284	3.1%
7	Technology	Web Developer	Los Angeles	\$82,970	3.4%
8	Technology	Web Developer	Houston	\$72,252	2.0%
9	Technology	Web Developer	Chicago	\$78,251	2.9%
10	Technology	Web Developer	Boston	\$85,334	3.2%
11	Technology	Web Developer	Atlanta	\$77,856	2.7%
12	Technology	Web Designer	Washington DC	\$69,062	0.3%
13	Technology	Web Designer	Seattle		
14	Technology	Web Designer	San Francisco		
15	Technology	Web Designer	Philadelphia		
16	Technology	Web Designer	New York City		
17	Technology	Web Designer	Los Angeles		
18	Technology	Web Designer	Houston		
19	Technology	Web Designer	Chicago		
20	Technology	Web Designer	Boston		
21	Technology	Web Designer	Atlanta		

Field	Description
Industry	The name of the industry, or job category, for each job title (17 total)
Job Title	The name of the job title (84 total)
City	The name of US cities where these job titles can be found (10 total)
Average Salary	The average salary in 2020 for each job title in each city
YoY % Growth	The percentage change in average salary from 2019 to 2020 for each job title in each city

DYNAMIC EXCEL

Dynamic Excel

Spill Ranges

SORT & SORTBY

FILTER

UNIQUE

SEQUENCE

RANDARRAY

Legacy Functions

Excel's calculation engine has changed, and **formulas will never be the same**

BEFORE:

ONE formula = **ONE** value

	A	B	C	D	E
1		Oct	Nov	Dec	4th Qtr.
2	Products	15%	35%	50%	Sales Target
3	Smart Speaker	\$3,420			\$22,800
4	Baseball Bat				\$1,080
5	Cowboy Boots				\$2,640
6	Vinyl Records				\$1,800
7	Violin				\$6,000
8	Sunglasses				\$7,440
9	Blu-Ray Player				\$12,000
10	Tennis Racket				\$840

"Legacy" Excel ↗

NOW:

ONE formula = **MANY** values

	A	B	C	D	E
1		Oct	Nov	Dec	4th Qtr.
2	Products	15%	35%	50%	Sales Target
3	Smart Speaker	\$3,420	\$7,980	\$11,400	\$22,800
4	Baseball Bat	\$162	\$378	\$540	\$1,080
5	Cowboy Boots	\$396	\$924	\$1,320	\$2,640
6	Vinyl Records	\$270	\$630	\$900	\$1,800
7	Violin	\$900	\$2,100	\$3,000	\$6,000
8	Sunglasses	\$1,116	\$2,604	\$3,720	\$7,440
9	Blu-Ray Player	\$1,800	\$4,200	\$6,000	\$12,000
10	Tennis Racket	\$126	\$294	\$420	\$840

"Dynamic" Excel ↗

DYNAMIC EXCEL

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Legacy Functions



A single formula in a single cell can return **many** results

- Values "spill" across adjacent cells
- Results are not hard-coded, and can change dynamically with the source data



New **dynamic array functions** are now available

- Functions like SORT, FILTER and UNIQUE take advantage of the new calculation engine
- Combining dynamic array functions can unlock brand new capabilities in Excel



Dynamic array behavior applies to **traditional Excel formulas** as well

- Existing functions can also leverage Excel's new calculation engine



Array functions are now **simpler and easier to learn**

- Traditional CSE array functions were typically only used by "experts"
- New dynamic array formulas are intuitive and user-friendly

DYNAMIC EXCEL

Dynamic Excel

Spill Ranges

SORT & SORTBY

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UNIQUE

SEQUENCE

RANDARRAY

Legacy Functions

Formulas that can return arrays of variable size are called **dynamic arrays**; these formulas are entered in a single cell and can “spill” results across an entire range

	A	B	C	D
1	Product	Sales	Margin	Profit
2	Smart Speaker	\$19,000	20%	
3	Baseball Bat	\$900	5%	
4	Cowboy Boots	\$2,200	15%	
5	Vinyl Records	\$1,500	10%	
6	Lego Bricks	\$5,000	20%	
7	Sunglasses	\$6,200	20%	
8	Blu-Ray Player	\$10,000	35%	
9	Tennis Racket	\$700	25%	
10	Leather Jacket	\$8,000	15%	

✕ ✓ *f_x* **=B2:B10*C2:C10**

	A	B	C	D
1	Product	Sales	Margin	Profit
2	Smart Speaker	\$19,000	20%	\$3,800
3	Baseball Bat	\$900	5%	\$45
4	Cowboy Boots	\$2,200	15%	\$330
5	Vinyl Records	\$1,500	10%	\$150
6	Lego Bricks	\$5,000	20%	\$1,000
7	Sunglasses	\$6,200	20%	\$1,240
8	Blu-Ray Player	\$10,000	35%	\$3,500
9	Tennis Racket	\$700	25%	\$175
10	Leather Jacket	\$8,000	15%	\$1,200

The resulting range of cells is known as the **spill range**



HEY THIS IS IMPORTANT!

The formula itself only lives in the **first cell** of the spilled range

SPILL RANGE PROPERTIES

Dynamic Excel

Spill Ranges

SORT & SORTBY

FILTER

UNIQUE

SEQUENCE

RANDARRAY

Legacy Functions

The **spill range** contains the results of a single dynamic array formula

E4					
					<i>f_x</i> =C2:C10*D2:D10
1	Product	Category	Sales	Margin	Profit
2	Smart Speaker	Electronics	\$19,000	20%	\$3,800
3	Baseball Bat	Sports	\$900	5%	\$45
4	Cowboy Boots	Clothing	\$2,200	15%	\$330
5	Vinyl Records	Music	\$1,500	10%	\$150
6	Lego Bricks	Games	\$5,000	20%	\$1,000
7	Sunglasses	Clothing	\$6,200	20%	\$1,240
8	Blu-Ray Player	Electronics	\$10,000	35%	\$3,500
9	Tennis Racket	Sports	\$700	25%	\$175
10	Leather Jacket	Clothing	\$8,000	15%	\$1,200
11					
12		Total Sales:			
13		Total Profit:	=SUM(E2#)		

Only the **first cell** in the range is **editable** (all others within the spill range are grayed out)

Cell formatting isn't carried over from the source and **does not spill**

Spill ranges are highlighted with a **blue border** when selected

Spilled ranges **update automatically** and **resize** to fit the resulting array

The **"#"** symbol can be used to **reference an entire spilled range**

#SPILL! ERRORS

Dynamic Excel

Spill Ranges

SORT & SORTBY

FILTER

UNIQUE

SEQUENCE

RANDARRAY

Legacy Functions

#SPILL! errors occur when something is “blocking” the spill range, since Excel will not overwrite existing values by default

- Clear any existing text, values, or merged cells from the entire spill range to fix the error

D2					
	A	B	C	D	E
1	Product	Sales	Margin	Profit	
2	Smart Speaker	\$19,000	20%	#SPILL!	
3	Baseball Bat	\$900	5%		
4	Cowboy Boots	\$2,200	15%		
5	Vinyl Records	\$1,500	10%		
6	Lego Bricks	\$5,000	20%		
7	Sunglasses	\$6,200	20%		
8	Blu-Ray Player	\$10,000	35%		
9	Tennis Racket	\$700	25%		
10	Leather Jacket	\$8,000	15%		



HEY THIS IS IMPORTANT!

You may also see a #SPILL! error if you try to use dynamic array formulas **inside of a table**

PRO TIP: GROWING SOURCE DATA

References within dynamic array formulas **do not automatically resize** as you add new data, but there are several ways to accommodate dynamic source data:

- 1 Format the data as a **table** and use structured references

	A	B	C	D	E	F	G
1	Product	Category	Sales	Margin		Categories	
2	Smart Speaker	Electronics	\$19,000	20%		=UNIQUE(Table1[Category])	
3	Baseball Bat	Sports	\$900	5%			
4	Cowboy Boots	Clothing	\$2,200	15%			
5	Vinyl Records	Music	\$1,500	10%			
6	Violin	Music	\$5,000	20%			
7	Sunglasses	Clothing	\$6,200	20%			
8	Blu-Ray Player	Electronics	\$10,000	35%			
9	Tennis Racket	Sports	\$700	25%			

Any new rows will be included

- 2 Select the **extra rows** to accommodate new records

	A	B	C	D	E	F	G
1	Product	Category	Sales	Margin		Categories	
2	Smart Speaker	Electronics	\$19,000	20%		=UNIQUE(B2:B16)	
3	Baseball Bat	Sports	\$900	5%			
4	Cowboy Boots	Clothing	\$2,200	15%			
5	Vinyl Records	Music	\$1,500	10%			
6	Violin	Music	\$5,000	20%			
7	Sunglasses	Clothing	\$6,200	20%			
8	Blu-Ray Player	Electronics	\$10,000	35%			
9	Tennis Racket	Sports	\$700	25%			
10							
11							
12							
13							
14							
15							
16							

These rows will be included (+ blanks)

- 3 Define a dynamic **named range** using OFFSET & COUNTA

Name Manager			?	X
New...	Edit...	Delete	Filter	
Name	Value	Refers To		
DynamicRange	(...)	=OFFSET(Sheet1!\$B\$2,0,0,COUNTA(Sheet1!\$B:\$B)-1)		
This named range will grow as new items are added to column B				
Refers to: =OFFSET(Sheet1!\$B\$2,0,0,COUNTA(Sheet1!\$B:\$B)-1)				
Close				

DYNAMIC ARRAY FUNCTIONS

SORT()

Sorts an array of data by one or more columns in the array

SORTBY()

Sorts an array of data by one or more columns in another array

FILTER()

Filters an array of data based on specified criteria and returns the matching records

UNIQUE()

Removes duplicates from an array of data and returns the unique records

SEQUENCE()

Generates a one- or two-dimensional array of sequential numbers

RANDARRAY()

Generates a one- or two-dimensional array of random numbers

SORT

SORT()

Sorts an array of data by one or more columns in the array

=SORT (array, [sort_index], [sort_order], [by_col])

An array of cells
that you want to
sort

Column # you want
to sort by
(Default is 1)

1 = Ascending
-1 = Descending
(Default is 1)

TRUE/1 = Sort by column
FALSE/0 = Sort by row
(Default is FALSE or 0)

F2				=SORT(A2:D10,4,-1)					
	A	B	C	D	E	F	G	H	I
1	Product	Sales	Margin	Profit		Product	Sales	Margin	Profit
2	Smart Speaker	\$19,000	20%	\$3,800		Smart Speaker	19000	0.2	3800
3	Baseball Bat	\$900	5%	\$45		Blu-Ray Player	10000	0.35	3500
4	Cowboy Boots	\$2,200	15%	\$330		Sunglasses	6200	0.2	1240
5	Vinyl Records	\$1,500	10%	\$150		Leather Jacket	8000	0.15	1200
6	Lego Bricks	\$5,000	20%	\$1,000		Lego Bricks	5000	0.2	1000
7	Sunglasses	\$6,200	20%	\$1,240		Cowboy Boots	2200	0.15	330
8	Blu-Ray Player	\$10,000	35%	\$3,500		Tennis Racket	700	0.25	175
9	Tennis Racket	\$700	25%	\$175		Vinyl Records	1500	0.1	150
10	Leather Jacket	\$8,000	15%	\$1,200		Baseball Bat	900	0.05	45

The array in A2:D10 is being
sorted by the 4th column
(Profit) in **descending** order

SORT

SORT()

Sorts an array of data by one or more columns in the array

```
=SORT ( array, [sort_index], [sort_order], [by_col] )
```

*An array of cells
that you want to
sort*

Column # you want
to sort by
(Default is 1)

1 = Ascending
-1 = Descending
(Default is 1)

TRUE/1 = Sort by column
FALSE/0 = Sort by row
(Default is FALSE or 0)

	A	B	C	D	E	F	G	H	I
1	Product	Sales	Margin			Product	Sales	Margin	Profit
2	Smart Speaker	\$19,000	20%	\$3,800		Baseball Bat	900	0.05	45
3	Leather Jacket	\$8,000	20%	\$1,600		Vinyl Records	1500	0.1	150
4	Sunglasses	\$6,200	20%	\$1,240		Leather Jacket	8000	0.15	1200
5	Vinyl Records	\$1,900	10%	\$190		Cowboy Boots	2200	0.15	330
6	Lego Bricks	\$5,000	20%	\$1,000		Smart Speaker	19000	0.2	3800
7	Junglasses	\$6,200	20%	\$1,240		Sunglasses	6200	0.2	1240
8	Blu-Ray Player	\$10,000	35%	\$3,500		Lego Bricks	5000	0.2	1000
9	Tennis Racket	\$700	25%	\$175		Tennis Racket	700	0.25	175
10	Leather Jacket	\$8,000	15%	\$1,200		Blu-Ray Player	10000	0.35	3500

The array in **A2:D10** is being sorted by the **3rd column** (Margin) in **ascending** order, then the **4th column** (Profit) in **descending** order

SORTBY

SORTBY()

Sorts an array of data by one or more columns in another array

=**SORTBY** (array, by_array, [sort_order], [array/order], [...])

*An array of cells that
you want to sort*

*Array of cells that
you want to sort by*

1 = Ascending
-1 = Descending
 (Default is 1)

*Additional pairs of
arrays to sort by*

F2							

HEY THIS IS IMPORTANT!

The array you sort by must be the **same size** as the array you are sorting

This array is sorted by **Profit** in **descending** order (even though it isn't in the array!)

FILTER

FILTER()

Filters an array of data based on specified criteria and returns the matching records

=FILTER (array, include, [if_empty])

An array of cells that you want to filter

A logical test to determine the filter criteria, where values of **TRUE** will be kept

An optional value to return if nothing passes the filter criteria

E4							
	A	B	C	D	E	F	G
1	Product	Category	Sales		Category: Clothing		
2	Smart Speaker	Electronics	\$19,000				
3	Baseball Bat	Sports	\$900				
4	Cowboy Boots	Clothing	\$2,200				
5	Vinyl Records	Music	\$1,500				
6	Lego Bricks	Games	\$5,000				
7	Sunglasses	Clothing	\$6,200				
8	Blu-Ray Player	Electronics	\$10,000				
9	Tennis Racket	Sports	\$700				
10	Leather Jacket	Clothing	\$8,000				

Product	Category	Sales
Cowboy Boots	Clothing	2200
Sunglasses	Clothing	6200
Leather Jacket	Clothing	8000

This array returns values from **A2:C10**, where Category = **Clothing**

FILTER

FILTER()

Filters an array of data based on specified criteria and returns the matching records

=FILTER (array, include, [if_empty])

An array of cells that you want to filter

A logical test to determine the filter criteria, where values of **TRUE** will be kept

An optional value to return if nothing passes the filter criteria

E5							
	A	B	C	D	E	F	G
1	Product	Category	Sales		Category: Clothing		
2	Smart Speaker	Electronics	\$19,000		Sales: \$5,000		
3	Baseball Bat	Sports	\$900				
4	Cowboy Boots	Clothing	\$2,200				
5	Vinyl Records	Music	\$1,500				
6	Lego Bricks	Games	\$5,000				
7	Sunglasses	Clothing	\$6,200				
8	Blu-Ray Player	Electronics	\$10,000				
9	Tennis Racket	Sports	\$700				
10	Leather Jacket	Clothing	\$8,000				

Product	Category	Sales
Sunglasses	Clothing	6200
Leather Jacket	Clothing	8000

To create an **AND** condition between multiple logical tests, you can **multiply** them together

This array returns values from **A2:C10** where Category = **Clothing** AND Sales > **5,000** (BOTH criteria must be met)

FILTER

FILTER()

Filters an array of data based on specified criteria and returns the matching records

=FILTER (array, include, [if_empty])

An array of cells that you want to filter

A logical test to determine the filter criteria, where values of **TRUE** will be kept

An optional value to return if nothing passes the filter criteria

E5 **=FILTER(A2:C10,(B2:B10=F1)+(C2:C10>F2),"No results")**

	A	B	C	D	E	F	G
1	Product	Category	Sales		Category:	Clothing	
2	Smart Speaker	Electronics	\$19,000		Sales:	\$5,000	
3	Baseball Bat	Sports	\$900				
4	Cowboy Boots	Clothing	\$2,200				
5	Vinyl Records	Music	\$1,500				
6	Lego Bricks	Games	\$5,000				
7	Sunglasses	Clothing	\$6,200				
8	Blu-Ray Player	Electronics	\$10,000				
9	Tennis Racket	Sports	\$700				
10	Leather Jacket	Clothing	\$8,000				

	Product	Category	Sales
1	Smart Speaker	Electronics	19000
2	Cowboy Boots	Clothing	2200
3	Sunglasses	Clothing	6200
4	Blu-Ray Player	Electronics	10000
5	Leather Jacket	Clothing	8000

To create an **OR** condition between multiple logical tests, you can **sum** them together

This array returns values from **A2:C10** where
Category = **Clothing** OR Sales > **5,000**
(EITHER criteria must be met)

UNIQUE

UNIQUE()

Removes duplicates from an array of data and returns only the unique records

=UNIQUE (array, [by_col], [exactly_once])

An array of cells that you want to remove duplicates from

TRUE/1 = Remove duplicates in columns
FALSE/0 = Remove duplicates in rows
(Default is FALSE or 0)

TRUE/1 = Extract values that **only appear once**
FALSE/0 = Extract all unique values
(Default is FALSE or 0)

G2 **=UNIQUE(B2:B10)**

	A	B	C	D	E	F	G
1	Product	Category	Sales	Margin	Profit		Categories
2	Smart Speaker	Electronics	\$19,000	20%	\$3,800		Electronics
3	Baseball Bat	Sports	\$900	5%	\$45		Sports
4	Cowboy Boots	Clothing	\$2,200	15%	\$330		Clothing
5	Vinyl Records	Music	\$1,500	10%	\$150		Music
6	Lego Bricks	Games	\$5,000	20%	\$1,000		Games
7	Sunglasses	Clothing	\$6,200	20%	\$1,240		
8	Blu-Ray Player	Electronics	\$10,000	35%	\$3,500		
9	Tennis Racket	Sports	\$700	25%	\$175		
10	Leather Jacket	Clothing	\$8,000	15%	\$1,200		

This array returns the unique Category values from **B2:B10**

UNIQUE

UNIQUE()

Removes duplicates from an array of data and returns only the unique records

=UNIQUE (array, [by_col], [exactly_once])

An array of cells that you want to remove duplicates from

TRUE/1 = Remove duplicates in columns
FALSE/0 = Remove duplicates in rows
(Default is FALSE or 0)

TRUE/1 = Extract values that **only appear once**
FALSE/0 = Extract all unique values
(Default is FALSE or 0)

G2						
=UNIQUE(B2:B10,FALSE,TRUE)						
1	Product	Category	Sales	Margin	Profit	Categories
2	Smart Speaker	Electronics	\$19,000	20%	\$3,800	Music
3	Baseball Bat	Sports	\$900	5%	\$45	Games
4	Cowboy Boots	Clothing	\$2,200	15%	\$330	
5	Vinyl Records	Music	\$1,500	10%	\$150	
6	Lego Bricks	Games	\$5,000	20%	\$1,000	
7	Sunglasses	Clothing	\$6,200	20%	\$1,240	
8	Blu-Ray Player	Electronics	\$10,000	35%	\$3,500	
9	Tennis Racket	Sports	\$700	25%	\$175	
10	Leather Jacket	Clothing	\$8,000	15%	\$1,200	

This array returns the Category values from B2:B10 that **appear exactly once**



PRO TIP: Include multiple columns in the array to return each unique **combination** of values

PRO TIP: Combining SORT, FILTER & UNIQUE

You can **combine**, or “nest”, multiple dynamic array functions to perform multiple operations

F2						
=SORT(UNIQUE(B2:B10))						
1	Product	Category	Sales	Margin	Categories	
2	Smart Speaker	Electronics	\$19,000	20%	Clothing	
3	Baseball Bat	Sports	\$900	5%	Electronics	
4	Cowboy Boots	Clothing	\$2,200	15%	Games	
5	Vinyl Records	Music	\$1,500	10%	Music	
6	Lego Bricks	Games	\$5,000	20%	Sports	
7	Sunglasses	Clothing	\$6,200	20%		
8	Blu-Ray Player	Electronics				
9	Tennis Racket	Sports				
10	Leather Jacket	Clothing				

Here we're combining **SORT** and **UNIQUE** to return an array of unique categories in ascending order

E4						
=SORT(FILTER(A2:C10,B2:B10=F1),3,-1)						
1	Product	Category	Sales	Category: Clothing		
2	Smart Speaker	Electronics	\$19,000			
3	Baseball Bat	Sports	\$900			
4	Cowboy Boots	Clothing	\$2,200			
5	Vinyl Records	Music	\$1,500			
6	Lego Bricks	Games	\$5,000			
7	Sunglasses	Clothing	\$6,200			
8	Blu-Ray Player	Electronics	\$10,000			
9	Tennis Racket	Sports	\$700			
10	Leather Jacket	Clothing	\$8,000			

Product	Category	Sales
Leather Jacket	Clothing	8000
Sunglasses	Clothing	6200
Cowboy Boots	Clothing	2200

Here we're combining **SORT** and **FILTER** to return an array of products in the Clothing category, sorted by Sales

SEQUENCE

SEQUENCE()

Generates a one- or two-dimensional array of sequential numbers

=SEQUENCE (rows, [columns], [start], [step])

Number of
rows to return

Number of
columns to return

Starting number
(Default is 1)

Increment between each number
(Default is 1)

	A	B	C	D	E	F
1	10	15	20	25	30	35
2	40	45	50	55	60	65
3	70	75	80	85	90	95
4	100	105	110	115	120	125
5	130	135	140	145	150	155
6	160	165	170	175	180	185
7	190	195	200	205	210	215
8	220	225	230	235	240	245
9	250	255	260	265	270	275
10	280	285	290	295	300	305



PRO TIP: Nest **SEQUENCE** within other functions to make them more dynamic

This generates a **10-row, 6-column** array starting at **10** and incrementing by **5** (note the numbers go left-to-right, then down)

RANDARRAY

RANDARRAY()

Generates a one- or two-dimensional array of random numbers

=RANDARRAY ([rows], [columns], [min], [max], [integer])

Number of
rows to return
(Default is 1)

Number of
columns to return
(Default is 1)

Minimum
value to return
(Default is 0)

Maximum
value to return
(Default is 1)

Return whole
numbers?
(Default is FALSE or 0)

	A	B	C	D	E	F	G
1	55	68	21	64	23	48	74
2	84	88	5	94	28	91	44
3	67	90	81	0	93	4	70
4	28	36	25	4	96	50	73
5	91	93	69	100	77	96	89
6	78	25	66	57	35	90	18
7	76	76	61	1	79	7	79
8	98	69	36	22	91	47	66
9	33	86	2	3	22	84	68
10	67	72	67	78	33	40	69



PRO TIP: Use **RANDARRAY** to randomly sort lists of data

This generates a **10-row by 7-column** array of random **whole numbers** between **0** and **100**

LEGACY FUNCTION: FREQUENCY

FREQUENCY()

Returns the frequency of values in a range based on specified intervals (or bins)

=FREQUENCY (data_array, bins_array)

An array of cells
containing values

An array of intervals (bins)
for grouping the values



HEY THIS IS IMPORTANT!

The **bins_array** argument must include the upper limit of each bin, formatted as a number

G2							=FREQUENCY(C2:C10,F2:F6)	
	A	B	C	D	E	F	G	
1	Product	Category	Sales		Bin	Upper Limit	Frequency	
2	Smart Speaker	Electronics	\$19,000		<= \$3,000	\$3,000	4	
3	Baseball Bat	Sports	\$900		\$3,001-\$6,000	\$6,000	1	
4	Cowboy Boots	Clothing	\$2,200		\$6,001-\$9,000	\$9,000	2	
5	Vinyl Records	Music	\$1,500		\$9,001-\$12,000	\$12,000	1	
6	Lego Bricks	Games	\$5,000		\$12,001-\$15,000	\$15,000	0	
7	Sunglasses	Clothing	\$6,200		> \$15,000		1	
8	Blu-Ray Player	Electronics	\$10,000					
9	Tennis Racket	Sports	\$700					
10	Leather Jacket	Clothing	\$8,000					

Here we're counting the frequency of Sales records which fall into each bin in **F2:F6**

NOTE: FREQUENCY always returns **one extra row** to account for values above the largest defined bin

LEGACY FUNCTION: TRANSPOSE

TRANSPOSE()

Flips a vertical range of cells to a horizontal range, or vice versa

=TRANSPOSE (array)

An array of cells you
want to transpose



PRO TIP: Use **TRANSPOSE** to "pivot" your data by turning rows into columns, or vice versa

E1	=TRANSPOSE(A1:C9)												
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product	Category	Sales		Product	Smart Speaker	Baseball Bat	Cowboy Boots	Vinyl Records	Violin	Sunglasses	Blu-Ray Player	Tennis Racket
2	Smart Speaker	Electronics	\$19,000		Category	Electronics	Sports	Clothing	Music	Music	Clothing	Electronics	Sports
3	Baseball Bat	Sports	\$900		Sales	19000	900	2200	1500	5000	6200	10000	700
4	Cowboy Boots	Clothing	\$2,200										
5	Vinyl Records	Music	\$1,500										
6	Violin	Music	\$5,000										
7	Sunglasses	Clothing	\$6,200										
8	Blu-Ray Player	Electronics	\$10,000										
9	Tennis Racket	Sports	\$700										

Here we're flipping the original array in **A1:C9** by turning each column into a row

Here we're flipping the original array in **A1:C9** by turning each column into a row

PRO TIP: JOINING ARRAYS WITH CHOOSE

The **CHOOSE** function can be used to combine separate cell ranges into a single array, which can be referenced or manipulated within other formulas

	A	B	C	D
1	Product	Category	Sales	Margin
2	Smart Speaker	Electronics	\$19,000	20%
3	Baseball Bat	Sports	\$900	5%
4	Cowboy Boots	Clothing	\$2,200	15%
5	Vinyl Records	Music	\$1,500	10%
6	Lego Bricks	Games	\$5,000	20%
7	Sunglasses	Clothing	\$6,200	20%
8	Blu-Ray Player	Electronics	\$10,000	35%
9	Tennis Racket	Sports	\$700	25%
10	Leather Jacket	Clothing	\$8,000	15%

Raw data at the product-level

F	G
Category	Avg. Sales
Electronics	\$14,500
Sports	\$800
Clothing	\$5,467
Music	\$1,500
Games	\$5,000

Avg Sales by Category
(using **AVERAGEIF**)

Unique list of Categories
(using **UNIQUE**)

I	J
Category	Avg. Sales
Electronics	\$14,500
Sports	\$800
Clothing	\$5,467
Music	\$1,500
Games	\$5,000

=CHOOSE({1,2},F2#,G2#)



PRO TIP: Using the array constant **{1,2}** tells Excel to combine both referenced ranges into one dynamic array

PRO TIP: THE LET FUNCTION

LET()

Allows you to declare variables, assign values, and use them within formulas

=LET (name1, name_value1, calculation_or_name2, [name_value2], [...])

Name of the variable
(must begin with a letter)

Value or calculation
assigned to the variable

A calculation using the
variable, or the name of
another variable (optional)

Additional pairs of variable
names and values

	A	B	C	D	E	F
1	Product	Category	Sales	Margin	Profit	
2	Smart Speaker	Electronics	\$19,000	20%	3800	
3	Baseball Bat	Sports	\$900	5%	45	
4	Cowboy Boots	Clothing	\$2,200	15%	330	
5	Vinyl Records	Music	\$1,500	10%	150	
6	Violin	Music	\$5,000	20%	1000	
7	Sunglasses	Clothing	\$6,200	20%	1240	
8	Blu-Ray Player	Electronics	\$10,000	35%	3500	
9	Tennis Racket	Sports	\$700	25%	175	
10	Leather Jacket	Clothing	\$8,000	15%	1200	



PRO TIP: Use **LET** to write clean, efficient, user-friendly formulas

This defines two variables, **Sales** and **Margin**, and multiplies them to return the profit